Practical 6

Aim: To develop an application to consume **Google Maps RESTful Web Service**.

We will create a **simple web page that shows a location on Google Map**.

Example: **Show Mumbai on Google Map**

1️⃣ Software Required

- NetBeans IDE
- JDK 8
- GlassFish Server
- Internet connection

2️⃣ Create Web Application

Open **NetBeans**
Go to: File → New Project
Select: Java Web → Web Application
Project Name: GoogleMapApp
Server: GlassFish Server
Click **Finish**

3️⃣ Get Google Maps API Key

1. Go to https://console.cloud.google.com
2. Create a **new project**
3. Enable: Maps JavaScript API
4. Create **API Key**

Example: AIzaSyXXXXXXXXXXXX

(Students will replace with their own key)

4️⃣ Create JSP Page

Create file: map.jsp

5️⃣ JSP Code

Paste this code.

```
<html>
<head>
<title>Google Maps Example</title>
<script src="https://maps.googleapis.com/maps/api/js?key=YOUR_API_KEY"></script>
<script>
function initMap()
{
  var location = {lat:19.0760, lng:72.8777}; // Mumbai
  var map = new google.maps.Map(
    document.getElementById("map"),
    {
      zoom:10,
      center:location
    }
  );
  var marker = new google.maps.Marker(
  {
    position:location,
    map:map,
    title:"Mumbai"
  });
}
</script>
</head>
<body onload="initMap()">
<h2>Google Map Example</h2>
<div id="map" style="height:500px; width:600px;"></div>
</body>
</html>
```

6️⃣ Replace API Key

Replace this line: YOUR_API_KEY

Example: https://maps.googleapis.com/maps/api/js?key=AIzaSyXXXX

7️⃣ Run Application

Run project.

Open browser: http://localhost:8080/GoogleMapApp/map.jsp

Output

The page will display:

- Google Map
- Marker on **Mumbai**

Technologies Used

- **Google Maps REST API**
- **JavaScript**
- **JSP**
- **GlassFish Server**